

Mathematical modelling of infectious disease

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Mathematical models can project how infectious diseases progress to show the likely outcome of an epidemic and help inform public health interventions. Models use some basic assumptions and mathematics to find parameters for various infectious diseases and use those parameters to calculate the effects of possible interventions, like mass vaccination programmes.

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History

Early pioneers in infectious disease modelling were William Hamer^[1] and Ronald Ross, who in the early twentieth century applied the law of mass action to explain epidemic behaviour. Lowell Reed and Wade Hampton Frost developed the Reed–Frost epidemic model to describe the relationship between susceptible, infected and immune individuals in a population.

Concepts

R_0 , the basic reproduction number

The average number of other individuals each infected individual will infect in a population that has no immunity to the disease.

S

The proportion of the population who are susceptible to the disease (neither immune nor infected).

A

The average age at which the disease is contracted in a given population.

L

The average life expectancy in a given population.

Assumptions

Models are only as good as the assumptions on which they are based. If a model makes predictions which are out of line with observed results and the mathematics is correct, the initial assumptions must change to make the model useful.

- Rectangular and stationary age distribution, i.e., everybody in the population lives to age L and then dies, and for each age (up to L) there is the same number of people in the population. This is often well-

justified for developed countries where there is a low infant mortality and much of the population lives to the life expectancy.

- Homogeneous mixing of the population, i.e., individuals of the population under scrutiny assort and make contact at random and do not mix mostly in a smaller subgroup. This assumption is rarely justified because social structure is widespread. For example, most people in London only make contact with other Londoners. Further, within London then there are smaller subgroups, such as the Turkish community or teenagers (just to give two examples), who mix with each other more than people outside their group. However, homogeneous mixing is a standard assumption to make the mathematics tractable.

Endemic steady state

An infectious disease is said to be endemic when it can be sustained in a population without the need for external inputs. This means that, on average, each infected person is infecting *exactly* one other person (any more and the number of people infected will grow exponentially and there will be an epidemic, any less and the disease will die out). In mathematical terms, that is:

$$R_0 = 1.$$

The basic reproduction number (R_0) of the disease, assuming everyone is susceptible, multiplied by the proportion of the population that is actually susceptible (S) must be one (since those who are not susceptible do not feature in our calculations as they cannot contract the disease). Notice that this relation means that for a disease to be in the endemic steady state, the higher the basic reproduction number, the lower the proportion of the population susceptible must be, and vice versa.

Assume the rectangular stationary age distribution and let also the ages of infection have the same distribution for each birth year. Let the average age of infection be A , for instance when individuals younger than A are susceptible and those older than A are immune (or infectious). Then it can be shown by an easy argument that the proportion of the population that is susceptible is given by:

$$S = \frac{A}{L}.$$

But the mathematical definition of the endemic steady state can be rearranged to give:

$$S = \frac{1}{R_0}.$$

Therefore, due to the transitive property:

$$\frac{1}{R_0} = \frac{A}{L} \Rightarrow R_0 = \frac{L}{A}.$$

This provides a simple way to estimate the parameter R_0 using easily available data.

For a population with an exponential age distribution,

$$R_0 = 1 + \frac{L}{A}.$$

This allows for the basic reproduction number of a disease given A and L in either type of population distribution.

Infectious disease dynamics

Mathematical models need to integrate the increasing volume of data being generated on host-pathogen interactions. Many theoretical studies of the population dynamics, structure and evolution of infectious diseases of plants and animals, including humans, are concerned with this problem.

Research topics include:

- transmission, spread and control of infection
- epidemiological networks
- spatial epidemiology
- persistence of pathogens within hosts
- intra-host dynamics
- immuno-epidemiology
- virulence
- Strain (biology) structure and interactions
- antigenic shift
- phylodynamics
- pathogen population genetics
- evolution and spread of resistance
- role of host genetic factors
- statistical and mathematical tools and innovations
- role and identification of infection reservoirs

Mathematics of mass vaccination

If the proportion of the population that is immune exceeds the herd immunity level for the disease, then the disease can no longer persist in the population. Thus, if this level can be exceeded by vaccination, the disease can be eliminated. An example of this being successfully achieved worldwide is the global smallpox eradication, with the last wild case in 1977. The WHO is carrying out a similar vaccination campaign to eradicate polio.

The herd immunity level will be denoted q . Recall that, for a stable state:

$$R_0 \cdot S = 1.$$

S will be $(1 - q)$, since q is the proportion of the population that is immune and $q + S$ must equal one (since in this simplified model, everyone is either susceptible or immune). Then:

$$\begin{aligned} R_0 \cdot (1 - q) &= 1, \\ 1 - q &= \frac{1}{R_0}, \\ q &= 1 - \frac{1}{R_0}. \end{aligned}$$

Remember that this is the threshold level. If the proportion of immune individuals *exceeds* this level due to a mass vaccination programme, the disease will die out.

We have just calculated the **critical immunisation threshold** (denoted q_c). It is the minimum proportion of the population that must be immunised at birth (or close to birth) in order for the infection to die out in the population.

$$q_c = 1 - \frac{1}{R_0}$$

When mass vaccination cannot exceed the herd immunity

If the vaccine used is insufficiently effective or the required coverage cannot be reached (for example due to popular resistance), the programme may fail to exceed q_c . Such a programme can, however, disturb the balance of the infection without eliminating it, often causing unforeseen problems.

Suppose that a proportion of the population q (where $q < q_c$) is immunised at birth against an infection with $R_0 > 1$. The vaccination programme changes R_0 to R_q where

$$R_q = R_0(1 - q)$$

This change occurs simply because there are now fewer susceptibles in the population who can be infected. R_q is simply R_0 minus those that would normally be infected but that cannot be now since they are immune.

As a consequence of this lower basic reproduction number, the average age of infection A will also change to some new value A_q in those who have been left unvaccinated.

Recall the relation that linked R_0 , A and L . Assuming that life expectancy has not changed, now:

$$R_q = \frac{L}{A_q},$$

$$A_q = \frac{L}{R_q} = \frac{L}{R_0(1 - q)}.$$

But $R_0 = L/A$ so:

$$A_q = \frac{L}{(L/A)(1 - q)} = \frac{AL}{L(1 - q)} = \frac{A}{1 - q}.$$

Thus the vaccination programme will raise the average age of infection, another mathematical justification for a result that might have been intuitively obvious. Unvaccinated individuals now experience a reduced force of infection due to the presence of the vaccinated group.

However, it is important to consider this effect when vaccinating against diseases that are more severe in older people. A vaccination programme against such a disease that does not exceed q_c may cause more deaths and complications than there were before the programme was brought into force as individuals will be catching the disease later in life. These unforeseen outcomes of a vaccination programme are called **perverse effects**.

When mass vaccination exceeds the herd immunity

If a vaccination programme causes the proportion of immune individuals in a population to exceed the critical threshold for a significant length of time, transmission of the infectious disease in that population will stop. This is known as elimination of the infection and is different from eradication.

Elimination

Interruption of endemic transmission of an infectious disease, which occurs if each infected individual infects less than one other, is achieved by maintaining vaccination coverage to keep the proportion of immune individuals above the critical immunisation threshold.

Eradication

Reduction of infective organisms in the wild worldwide to zero. So far, this has only been achieved for smallpox and rinderpest. To get to eradication, elimination in all world regions must be achieved.

See also

- Compartmental models in epidemiology
- Critical community size
- Ecosystem model
- Epidemic model
- Force of infection
- Landscape epidemiology
- Next-generation matrix
- Risk factor
- Sexual network
- Transmission risks and rates
- 1947 New York City smallpox outbreak
- Cross-species transmission
- Corrupted Blood incident

References

1. Hamer, W. (1928). *Epidemiology Old and New*. London: Kegan Paul

Further reading

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External links

- Institute for Disease Modeling (<http://www.idmod.org/>)
- WHO Collaborating Centre for Geospatial Disease Modelling, University of Oxford (<http://www.map.ox.ac.uk/>)
- Institute for Emerging Infections, University of Oxford (<http://www.emdis.ox.ac.uk/>)
- Center for Infectious Disease Modeling and Analysis, Yale School of Public Health (<http://cidma.yale.edu/>)
- Center for Infectious Disease Dynamics, The Pennsylvania State University (<http://www.cidd.psu.edu/>)
- Cambridge Infectious Diseases (<http://www.infectiousdisease.cam.ac.uk/>)
- Centre for the Mathematical Modelling of Infectious Diseases (<http://cmmid.lshtm.ac.uk/>), London School of Hygiene & Tropical Medicine
- Journal of the Royal Society Interface (<http://www.pubs.royalsoc.ac.uk/interface>)
- Models of Infectious Disease Agent Study (<http://www.epimodels.org/>)
- Infectious Disease Modeling: Measles Virus (<http://apmonitor.com/wiki/index.php/Apps/MeaslesVirus>)
- Model-Builder (<http://model-builder.sourceforge.net/>): Interactive (GUI-based) software to build, simulate, and analyze ODE models.

- Tuberculosis Modelling and Analysis Consortium (TB MAC) (<http://tb-mac.org/>): Group focused on improving global Tuberculosis control by coordinating and promoting mathematical modelling and other quantitative research activities.
- GLEaMviz Simulator (<http://www.gleamviz.org/>): Enables simulation of emerging infectious diseases spreading across the world.
- STEM (<http://www.eclipse.org/stem/>): Open source framework for Epidemiological Modeling available through the Eclipse Foundation.

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